

Data Visualization Assignment

Problem

Your task is to investigate the given problem using an appropriate dataset and demonstrate how data visualization can be used to understand the problem, identify patterns, and derive meaningful insights.

The objective is **not simply to create graphs**. You are expected to understand the data, formulate questions, select suitable visualizations, interpret them, and explain your findings.

Instructions

1. Understand the Given Problem

Read the problem statement carefully and identify:

- What is the problem about?
- What information is required to investigate the problem?
- What variables may be relevant?
- What type of patterns or relationships could exist?

Before starting the implementation, write a brief description of the problem in your own words.

2. Find or Collect an Appropriate Dataset

Obtain a dataset relevant to the given problem.

The dataset may be obtained from:

- Kaggle
- Government/open-data websites
- Public datasets
- Research datasets
- APIs
- Other reliable sources
- Self-collected data, if appropriate

Mention the **exact source of the dataset**.

Do not select a dataset only because it is easy to visualize. It should contain sufficient information to investigate the given problem.

3. Understand Your Dataset

Before creating any visualization, examine the dataset.

Identify:

- Number of rows
- Number of columns
- Column names
- Data types
- Numerical variables
- Categorical variables
- Missing values
- Duplicate records
- Possible outliers
- Important variables related to the given problem

Prepare a brief description of the important attributes.

4. Clean and Prepare the Data

Perform the necessary data cleaning and preprocessing.

Depending on your dataset, this may include:

- Handling missing values
- Removing duplicate records
- Correcting data types
- Handling inconsistent values
- Formatting dates
- Removing unnecessary columns
- Handling inappropriate or extreme values

Do not perform unnecessary cleaning.

For every major cleaning operation, mention **what you changed and why**.

5. Formulate Analytical Questions

Based on the given problem and your dataset, formulate **at least 5 meaningful questions** that you want to investigate.

For example:

Problem: Investigate student academic performance.

Possible questions:

- Is attendance related to academic performance?
- Does study time affect marks?
- Which group of students performs better?
- What is the distribution of marks?
- Are there students with unusually high or low performance?

Your questions must help you investigate the given problem.

Avoid questions that only require simple counting unless the count itself is meaningful.

6. Select Appropriate Visualizations

For each analytical question, decide which visualization is most appropriate.

You must create **at least 5 visualizations using at least 4 different visualization types**.

You may use:

- Bar Chart
- Line Chart
- Histogram
- Scatter Plot
- Box Plot
- Violin Plot
- Count Plot
- Pie Chart
- Heatmap
- Area Chart
- Stacked Bar Chart
- Other appropriate visualization techniques

Do not use a visualization simply to increase the number of charts.

For every visualization, clearly explain:

Why did you choose this visualization for this particular question?

7. Create and Format the Visualizations

Every visualization should be properly presented.

Where applicable, include:

- Appropriate title
- X-axis label
- Y-axis label
- Legend
- Appropriate scale
- Readable labels
- Suitable formatting

The visualization should allow another person to understand the information without requiring an explanation of the code.

8. Interpret Every Visualization

For every visualization, provide the following:

Question:

What question is the visualization answering?

Visualization:

Insert the chart.

Reason:

Why was this visualization selected?

Observation:

What do you observe from the chart?

Insight:

What meaningful information can be derived from the observation?

Do not simply write:

“The graph shows the data.”

Explain what the data actually tells you.

9. Find Patterns and Relationships

After creating the visualizations, identify important patterns in the data.

Look for:

- Trends
- Differences between groups
- Relationships between variables
- Correlations

- Distributions
- Outliers
- Increasing/decreasing patterns
- Unusual observations
- Similarities and differences

Your objective is to move from:

Data → Visualization → Observation → Insight

10. Create One Bad Visualization

Select one of your analytical questions and create an intentionally **inappropriate or misleading visualization**.

For example:

- Use an unsuitable chart type.
- Use a misleading axis.
- Use an inappropriate scale.
- Use unnecessary 3D effects.
- Use excessive visual elements.
- Present the data in a way that makes comparison difficult.

Then explain:

1. What is wrong with the visualization?
2. Why is it inappropriate or misleading?
3. How can it be improved?
4. Create the corrected visualization.

11. Compare Two Visualization Techniques

Select one important question from your investigation.

Answer the same question using **two different visualization techniques**.

For example:

Question: How does the distribution of marks vary across groups?

Visualization 1: Box Plot

Visualization 2: Violin Plot

Compare both visualizations and explain:

- Which one communicates the information better?
- Why?
- What information is easier to understand from each?
- Which visualization would you recommend?

12. Identify the Most Important Findings

After completing the investigation, write **at least 5 major findings**.

Your findings should come from the visualizations and should be related to the original problem.

Focus on meaningful findings rather than simply repeating numbers.

13. Give a Final Conclusion

Write a short conclusion explaining:

- What you investigated
- What you discovered
- The most important pattern or relationship
- Which visualization was most useful
- Whether the visual analysis helped answer the original problem

Also mention at least **two limitations** of your investigation.

Minimum Requirements

Your final investigation must contain:

- 1 given problem/topic
- 1 relevant dataset
- Dataset understanding
- Data cleaning/preprocessing
- Minimum 5 analytical questions
- Minimum 5 visualizations
- Minimum 4 different visualization types
- Interpretation of every visualization
- Minimum 1 intentionally bad visualization
- Corrected version of the bad visualization
- 1 question answered using 2 different visualization techniques
- Minimum 5 major findings
- Final conclusion

- Minimum 2 limitations
- Source/reference of the dataset
- Source code/workbook

Important

Do not create visualizations first and then try to find something to say about them.

Follow this sequence:

Given Problem → Understand Data → Clean Data → Ask Questions → Select Visualization → Create Visualization → Observe → Interpret → Find Insights → Conclude

The marks will depend on your **understanding, reasoning, visualization selection, and interpretation**, not merely on whether the code produces a graph.